SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0510**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Analytical Problem Solving

Code of Conduct:

I T S Sai Lahari/192325137 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **T S SAI LAHARI**

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation
(a) Explain the purpose of grey level transformation in image enhancement.

Grey level transformation is used to modify the intensity values of an image to improve its visual appearance. It helps in:

- Enhancing image contrast
- Highlighting important features
- Improving brightness
- Preparing images for further processing such as segmentation and object detection

One common transformation is the **negative transformation**, which reverses brightness values.
Formula:

$$s = 255 - r$$

where

- r= original pixel value
- s= transformed pixel value

(b) Apply a negative transformation s=255-r to the pixel values:
r=[50,100,150,200]
Compute the transformed values.

Given

$$r = [50,100,150,200]$$

Using

$$s = 255 - r$$

Calculations
For $r = 50$

$$s = 255 - 50 = 205$$

For $r = 100$

$$s = 255 - 100 = 155$$

For $r = 150$

$$s = 255 - 150 = 105$$

For $r = 200$

$$s = 255 - 200 = 55$$

Final Answer

Original Pixel (r)	Transformed Pixel (s)
50	205
100	155
150	105
200	55

2. Log Transformation
(a) Explain how log transformation enhances low-intensity pixels.

Given

$$r = [50,100,150,200]$$

Using

$$s = 255 - r$$

Calculations
For $r = 50$

$$s = 255 - 50 = 205$$

For $r = 100$

$$s = 255 - 100 = 155$$

For $r = 150$

$$s = 255 - 150 = 105$$

For $r = 200$

$$s = 255 - 200 = 55$$

Final Answer

Original Pixel (r)	**Transformed Pixel (s)**
50	205
100	155
150	105
200	55

(b) Given s=clog(1+r), where c=10, compute s for:
r=[0,10,50,100]

Given

$$c = 10$$
$$r = [0,10,50,100]$$

Using natural logarithm.
Calculations
For $r = 0$

$$s = 10\ln(1) = 0$$

For $r = 10$

$$s = 10\ln(11)$$
$$= 10(2.3979) = 23.98$$

For $r = 50$

$$s = 10\ln(51)$$
$$= 10(3.9318) = 39.32$$

For $r = 100$

$$s = 10\ln(101)$$
$$= 10(4.6151) = 46.15$$

Final Answer

r	s
0	0.00
10	23.98
50	39.32
100	46.15

---

3. Histogram Processing
(a) Explain the role of histogram equalization in contrast enhancement.

Histogram equalization improves image contrast by redistributing pixel intensity values over the entire intensity range.
Advantages:

- Enhances low-contrast images
- Makes hidden details visible
- Produces a more uniform histogram
- Improves image quality

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	**2**
1	**2**
2	**4**
3	**8**

Compute the cumulative distribution and equalized intensity values.

Given histogram

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels

$$N = 2 + 2 + 4 + 8 = 16$$

Maximum intensity

$$L = 4$$

Step 1: Cumulative Frequency

Intensity	Frequency	Cumulative Frequency
0	2	2
1	2	4
2	4	8
3	8	16

Step 2: Cumulative Distribution Function (CDF)

$$CDF = \frac{\text{Cumulative Frequency}}{16}$$

Intensity	CDF
0	0.125
1	0.25
2	0.50
3	1.00

Step 3: Equalized Intensity
Formula

$$s = (L - 1) \times CDF$$

Since

$$L = 4$$
$$L - 1 = 3$$

Intensity	CDF	Equalized Value
0	0.125	0
1	0.25	1
2	0.50	2
3	1.00	3

(Rounded to nearest integer.)

---

4. Spatial Filtering (Smoothing)
(a) Explain the purpose of spatial filtering for smoothing.

Spatial smoothing reduces image noise and removes small intensity variations.
Applications:

- Noise removal
- Blurring
- Preprocessing before segmentation
- Improving image quality

The averaging filter replaces each pixel with the average of its neighboring pixels.

(b) Apply a 3×3 averaging filter to the following region:
10 20 30 20 30 40 30 40 50
Compute the new center pixel value.

Given matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Sum

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average

$$\frac{270}{9} = 30$$

Final Answer
New center pixel value

$$\boxed{30}$$

---

5. Spatial Filtering (Sharpening)
(a) Explain how sharpening enhances image details.

Sharpening emphasizes edges and fine details by enhancing rapid intensity changes.
It is useful for

- Edge detection
- Medical imaging
- Satellite imaging
- Object recognition

Laplacian filtering highlights regions with sudden intensity changes.

(b) Apply the Laplacian mask:
[0 -1 0}, {-1 4 -1], { 0 -1 0]
to the matrix:
10 10 10, 10 20 10, 10 10 10
Compute the output at the center.

Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Center calculation

$$\begin{aligned} &= (4 \times 20) - (10 + 10 + 10 + 10) \\ &= 80 - 40 \\ &= 40 \end{aligned}$$

Final Answer

Output at center pixel

$$\boxed{40}$$